

SYLLABUS

National school on artificial intelligence

Level: Third year

Semester: S5

University year: 2025/2026

Identification of teaching module

Module: Stochastic Modeling and Simulation

Teaching Unit: Methodological

Number of Credits: 5

Coefficient: 3

Total number of hours per week:

o Classes: 2H, Tutorials: 1h30 and Labs: 30 minutes

Responsable of teaching module

Instructor: Dr. Nawel Arrar Remita, Associate Professor

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Description of the teaching module

Prerequisites: Probability, statistic, inference and computational tools to implementation.

General Objective: This module is designed to introduce you to the fundamental concepts of random systems modeling and the simulation techniques used to study them. At the end of this module, you will be able to:

- Understand and model stochastic dynamic systems using Markov chains.
- Use simulation methods to analyze complex random phenomena.
- Apply advanced techniques such as Monte Carlo and its variants to solve practical problems.

Learning Objectives: This course is divided into two parts: the first part is dedicated to stochastic modeling, with a review of Markov chains on discrete and continuous spaces and their matrix interpretations. The second part is dedicated to stochastic simulation, giving an overview of techniques for generating and evolving sequences of pseudo-random numbers, and their use in the production of random variables and vectors obeying any law. Another stochastic simulation idea, called the Monte Carlo method, is dedicated to solving purely deterministic problems, such as the evaluation of multiple integrals or combinatorial optimization. We also present classical Markov Chain Monte Carlo algorithms MCMC.

Course Content

Part 1: Stochastic Processes and their Classification

- o Discrete time Markov chains
- o Continuous time Markov chains

Part 2: Introduction to Stochastic Simulation

- o Pseudo random number generator
- o Simulation of random variables
- o Simulation of random vectors
- o Monte Carlo methods and variance reduction methods
- o Simulation of stochastic processes (MCMC)

Assessment methods

Nature du contrôle	Pondération en %
Final exam	60
Tutorials (Assignement + Attendance)	16
Lab	4
Mid term exam	20
Total	100%

References & Bibliography

Title of the work	Autor	Publisher
-Précis de recherche opérationnelle	-R. Faure, B. Lemaire, C. Picouleau	DUNOD
-Modélisation et simulation	-Amar Aissani	OPU
-Recherche opérationnelle pour Ingénieurs 2	-H. F. Hêche	Presses Polytechniques
-An Introduction to Stochastic Modeling.	-M. A. Pinsky et S. Karlin	4th ed. Academic Press

Course Schedule

Week	Titre du Cours
1	Part 1: Stochastic Processes and their Classification Chapter 1 Discret-Time Markov Chains - o Stochastic process - o Markov processes and Markov property - o Discret-Time Markov Chains - o Representative graphs, - o Probabilities and Transition Matrix - o
2	- o States and chains classification - o Stationary distribution - o Examples
3	Chapter 2 Continuous-Time Markov Chains - o Continuous-Time Markov Chains, - o Exponential distribution - o Poisson Processes - o Particular Markov Process : Birth and death processes - o Introduction to Queueing System 1. Queueing Model of type M/M/1
4	Performance Measures 2. Queueing Model of type M/M/s 3. Queueing Model of type M/M/s/K 4. Queueing Systems attended by a finite number of customers
5	o Other Models
6	- o Particular Queueing Systems - Markovian models - o Queueing Networks
7	Part 2: Introduction to Stochastic Simulation Chapter 3 Simulation of random variables Generating of Random Number - o Pseudo random number generator - o Congruential methods - o Von Neumann square method - o Validation of pseudo-random numbers Simulation of random variables - o Inverse method

8	Midterm exam
9	- o Accept-Reject method - o Composition method Specific methods
10	Simulation of Random Vectors - o Generating multivariate Gaussian vectors Random Process Generating - o DTMC Simulation & CTMC Simulation
11	Chapter 4 Monte Carlo Methods and Variance Reduction Methods Monte Carlo methods - o Description of the method - o Integration by Monte Carlo method - o Convergence and limit of the method
12	Variance Reduction Methods - o Preferential sampling or importance function - o Control variable
13	- o Antithetic variable - o Stratification method - o Average value or conditioning
14	Chapter 5 Simulation of stochastic processes (MCMC) - o Metropolis algorithm - o Generalisation of Metropolis-Hastings - o Boltzman-Gibbs algorithm
15	Part 3: Probabilistic Machine Learning Chapter 6 Bayesian Networks Hidden Markov chains - o Bayesian description - o Algorithm
16	Exams